

Bridge Day 16 INSTRUCTOR KEY

Exam 2 Rehearsal: Loops, Lists, Mutation, and Filtering

Learning sequence: Predict - Trace - Explain - Test - Interpret

Monday, July 27, 2026 • 20 points • target 17 minutes

Name		UIN		Section	
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Assessment boundary and evidence source

Contract: 07a04826c975

Cumulative foundations: input conversion, nested condition tracing, break/continue, indexed mutation, compound list filtering

Scaffold level: minimal-scaffold Exam 2 rehearsal

Evidence source: the matching v5 workbook readiness/evidence pages and VIP activities 18.25, 18.26, 18.27.

Show intermediate state for trace credit. Program credit requires one coherent solution, a stated assumption when the prompt leaves one implicit, and at least one test whose expected result can distinguish correct from incorrect logic.

Item 1. Trace skip, running state, collection, and early stop. - 6 points

```
values = [3, -2, 5, 0]
result = []
running = 0

for index in range(len(values)):
    if values[index] < 0:
        continue
    running = running + values[index]
    if running > 6:
        result.append(running)
        break
    result.append(values[index] * 2)

print(result, running, index)
```

Question: Give a complete reached-pass ledger, including skipped statements and the exact final output.

Model response: Index 0 adds 3 and appends 6. Index 1 is negative, so continue skips both updates. Index 2 adds 5, making running 8; the > 6 branch appends 8 and break ends the loop before index 3. Exact output: [6, 8] 8 2.

Item 2. Diagnose final-list evidence that disagrees with the total. - 4 points

```
values = [4, -3, 7, -1]
total = 0

for value in values:
    if value < 0:
        value = 0
    total = total + value

print(values, total)
```

Question: Give actual output, explain the mismatch, and write an indexed repair that makes the list and total describe the same final state.

Model response: Temporary value becomes 0 on negative passes, so total is 11, but the list remains [4, -3, 7, -1]. Repair with an index loop that assigns values[index] = 0 before adding values[index]. The repaired final list is [4, 0, 7, 0] and total remains 11.

Complete program - 10 points

- Read threshold as an integer and use values = [12, -4, 7, 20, 7].
- Loop by index and replace negative values with 0 before any summary.
- Append a final value to selected only when it is at least threshold and even.
- Compute the total of all final values and the index of the first largest final value without max or index.
- Print Final, Selected, Total, and Largest index on separate labeled lines. Give a threshold-equality test.

Program workspace

```
Model program
threshold = int(input())
values = [12, -4, 7, 20, 7]
selected = []
total = 0
largest = values[0]
largest_index = 0

for index in range(len(values)):
    if values[index] < 0:
        values[index] = 0
    value = values[index]
    total = total + value
    if value >= threshold and value % 2 == 0:
        selected.append(value)
    if value > largest:
        largest = value
        largest_index = index

print(f"Final: {values}")
print(f"Selected: {selected}")
print(f"Total: {total}")
print(f"Largest index: {largest_index}")
```

Test input, expected result, and why this test matters

Reasoning and test evidence All summaries follow mutation, so they describe one consistent final list. The compound filter requires both threshold evidence and evenness. Strict > preserves the first largest position. With threshold 8, final values are [12, 0, 7, 20, 7], selected is [12, 20], total is 46, and largest_index is 3. Threshold 20 must still select 20 because equality is included.

Scoring guidance: 1 input, 2 indexed mutation, 2 compound selection, 1 total, 2 first-largest tracking, 1 exact output, 1 equality test.